

Automated Driver Drowsiness Detection System using Computer Vision and Machine Learning

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Abstract— In recent years, road safety has become an increasingly pressing issue due to the increase in the number of fatal accidents throughout the world. Accidents are a major threat to road safety, and fatigue and drowsiness are the Primary causes of such incidents. In order to prevent such accidents, it is essential to detect and aware of signs that cause drowsiness or fatigue in a driver, and issue a timely warning. A range of technologies have been created to detect drowsy driving; however, computer vision and image processing technology can efficiently detect the drowsiness using facial expressions. As described, a system has been proposed that continuously monitors the driver's facial expression by analyzing the eyes and mouth movement in order to detect any signs of drowsiness or emotional shifts. If any such changes are detected, the system will alert the driver and helps in avoiding potential danger. This system provides a low-cost and resource-efficient approach to monitoring driver behavior and it utilizes facial landmark analysis to assess a driver's performance without the need for additional sensors or equipment.

Keywords— Road-Safety, Driver Drowsiness, Fatigue, Facial Expressions, Computer Vision, Image Processing, Alert.

I. INTRODUCTION

Road accidents in India continue to be a major cause of death and disability, accounting for 11% of all road-related deaths in the world. In 2019, there were 449,002 accidents resulting in 151,113 deaths and 451,361 injuries, a slight decrease from 2018[16]. The implementation of several measures that focus on road safety, is thought to have contributed to the decline. National Highways accounted for 35.7% of deaths, State Highways 24.8%, and other roads 39%. The majority of deaths (85%) were in the 18-60 age group. These accidents are often caused by an unrecognized factor - drowsy driving.

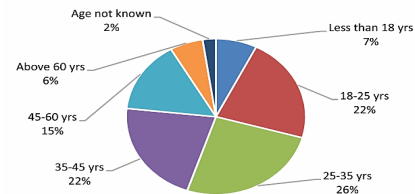


Fig. 1. Age profile of road accident deaths in 2019

What is Drowsy Driving?

Drowsy driving is when a driver operates a vehicle while being excessively tired or fatigued. It is a major safety concern on the roads, as it can lead to a lack of concentration and reaction time, as well as potentially dangerous driving behaviours. The effects of drowsy driving can be similar to those of drunk driving, often resulting in slower reaction times and impaired judgement [21]. Drivers who are feeling drowsy are more likely to make mistakes and fail to recognize potential hazards on the road, making them more susceptible to causing a crash. In addition to the potential for a fatal crash, drowsy driving can also result in costly property damage, as well as a variety of other serious injuries to both the driver and other people.

The National Highway Traffic Safety Administration estimates that one out of every five fatal crashes are caused by fatigued driving. According to the AAA Foundation for Traffic Safety, drowsy driving contributed to 72,000 crashes, 44,000 injuries, and 800 fatalities in 2013. According to the National Sleep Foundation, 37% of drivers and 60% of drivers have both driven while being sleepy in the last year. Drowsy driving is thought to cost the American economy \$109 billion annually.

In order to avoid drowsy driving and ensure driver safety, it is important for drivers to take the necessary precautions. This includes avoiding early-morning or late-night driving, obtaining adequate rest before driving, and taking frequent stops. Additionally, drivers should be aware

of the warning signs of drowsy driving, such as yawning, heavy eyelids, and difficulty keeping their eyes open [14].

Drivers can also take advantage of detection systems that are designed to alert them if they are feeling drowsy. These systems use sensors and artificial intelligence algorithms to detect signs of fatigue and alert the driver before anything happens. These technologies are designed to detect signs of drowsiness in drivers, such as yawning, blinking, and erratic steering, and alert them so that they can take corrective action. Some of the detection systems use a variety of sensors to monitor driver's behavior, such as video cameras and infrared sensors [18], and use artificial intelligence algorithms to detect signs of fatigue. If the system detects signs of drowsiness, it will alert the driver with a warning tone or vibration, as well as display a visual alert helping the drivers to reduce the risk of drowsy driving-related crashes.

In this paper, we explore the various models and methods used for drowsiness detection and propose the most suitable system. We evaluated different approaches to detect drowsiness, and found machine learning based approach with computer vision provide accurate results.

Our work compares various machine learning algorithms for detecting drowsiness and evaluate the effectiveness of each algorithm [8]. To determine the best suited for detecting drowsiness. The algorithms we evaluated include Decision Tree (DT), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve bayes, Random Forest and Logistic Regression. We analyse the performance of each algorithm in terms of accuracy and used it for training and testing.

The real-time detection technique makes use of the best trained classifier. A camera built within the system will record a live video of the driver's actions. The recorded footage is divided into separate frames. These frames are processed and used for facial landmarking. OpenCV and dlib can be used together to process images. The dlib tool kit is used to provides algorithms for facial recognition, facial Landmark detection, and face morphing [5]. It also provides a wide range of image processing functions, such as image filtering, image segmentation, and image registration. OpenCV can be used to detect objects in an image and dlib can be used to perform facial recognition. Additionally, OpenCV and dlib can be used together to perform image segmentation, image registration, and image filtering [17].

These images are used detect signs of drowsiness in drivers, such as yawning, blinking and to help drivers stay alert and safe on the road. This system does not include any additional sensors or external devices for detecting the driver's drowsiness.

The following is a breakdown of the paper's structure. In Section II, we defined various methods related to our work in detail. In Section III, we provide the methodology that we had worked on to develop a cost-effective and efficient model. In Section IV, the Experimental results have been provided for the proposed system. In Section V, the conclusion has described and finally, in Section VI, the future enhancement has been discussed.

II. RELATED WORKS

There are a variety of detection systems that can help ensure safe driving. These include:

Lane Departure Warning Systems: Y. Saito *et al.* [6] proposed a driver assistance system with a dual control system to detect driver fatigue and avoid lane deviation collisions. When a car drifts out of its lane, this system employs cameras and sensors to identify it and warn the driver with a tone or vibration.

Blind Spot Detection Systems: G. Liu *et al.* [9] proposed a radar-based blind spot detection and warning system for driver assistance. This system uses sensors to detect when another vehicle is in a driver's blind spot and alert the driver with a warning tone or vibration.

Adaptive Cruise Control Systems: C. Han *et al.* [7] proposed a paper on heterogeneous connected vehicle systems with strong cruise control. Low latency wireless vehicle-to-vehicle communication is offered by this system.

Respiratory Signal Analysis Systems: F. Guede-Fernández *et al.* [4] proposed a respiratory signal analysis-based system for detecting driver drowsiness. This method makes use of sensors to determine a driver's respiration rate, which is then utilized to determine how sleepy they are. The sensors analyze variations in the driver's respiratory rate to find indications of drowsiness.

Drowsiness Detection System Based on EEG Signals: M. Pathak *et al.* [15] aims at designing a single-channel EEG acquisition system for drowsiness detection that is economical and portable.

PERCLOS Method: K. S. Sankaran *et al.* [18] proposed an effective method named Percentage eye Closure (PERCLOS) that focuses on eye closure. The PERCLOS establishes a parameter level to detect the drowsiness. This method uses infrared or visible light sensors to measure the angle of the eyes and compare it to a predefined baseline.

Hybrid Machine Learning Algorithm: A. Altameem *et al.* [1] developed a hybrid machine learning algorithm for drowsiness detection that combines computer vision, artificial intelligence, and deep learning to detect signs of fatigue under different lighting situations. The advantages of using this approach for drowsiness detection such as its ability to detect fatigue earlier than traditional methods.

There are several problems associated with the existing driver drowsiness detection systems. Many existing methods use a single type of sensor or a single input signal which result inaccurate accuracy, limited applicability (work only in certain scenarios), requires to wear or use certain devices or sensors, lack of real world testing, and concerns in privacy.

Therefore, the development of an accurate, widely applicable, cost-effective, and ethically sound methods required.

III. METHODOLOGY

The proposed methodology is divided into two sections training a model and Real time monitoring.

A. Training a Model:

Training a model is the process of creating a machine learning algorithm that can accurately predict outcomes. In this we use various techniques and algorithms to create the model and optimize its performance.

The process consists of the following steps:

1) Collection of Data:

To develop an effective driver drowsiness detection model, we must first collect data on the various indicators of drowsiness, such as eye-closing, yawning, eye-opening, no-yawning and other physical signs of fatigue. Here we take a dataset containing 2900 images of 640 X 480 pixels classified into four different classes (eyes_closed, eyes_open, yawn, no_yawn). The images were taken under various conditions.



Fig. 2. Example for Yawning data



Fig. 3. Example for No- yawning data



Fig. 4. Example for opened-eye data



Fig. 5. Example for closed-eye data

2) Splitting the Data:

Data should be divided into two sets once it has been gathered: training data and testing data. In this model, the data is split into two portions: 90% is used to train the model, and 10% is used to test how well it performed. Finally, the model's performance is evaluated to determine how accurately it makes predictions.

The data and corresponding labels are loaded using the load_data function. The following table denotes the sample count of each class in the training and testing.

Labels	Sample Count
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Closed	726 images
Open	726 images
No_yawn	725 images
Yawn	723 images

TABLE I. Categorization of images

3) Data Analysis:

In this process, the training data is loaded and resized to 64x64 pixels using OpenCV. The resulting image arrays are saved as NumPy arrays. The images are resized and cropped to have a consistent format for extracting the features. Finally, the pixel intensity distribution of a randomly selected image is plotted using Matplotlib's histogram function and flattened into a 2D array.

4) Model Evaluation:

The next step after data analysis would be to define and train a machine learning model. This involves selecting an appropriate algorithm.

We used several algorithms to detect drowsiness, including K-Nearest Neighbors (KNN), Support Vector Machines (SVMs), Naive Bayes, Random Forests, Logistic Regression, and Decision Trees. Each algorithm was evaluated based on its ability to accurately detect drowsiness.

K-Nearest Neighbors (KNN): KNN is a non-parametric, lazy learning algorithm that can be applied to both regression and classification. KNN can be effective in detecting drowsiness since it takes into account the distance between points to determine the class of a given point. KNN can be used to generate a model that can detect drowsiness with a reasonable degree of accuracy. We trained and tested the model with the data and able to get 80.821%.

Naive Bayes: Naive Bayes is a probabilistic machine learning algorithm that uses Bayes' theorem for predicting the probability of a given class. Naive Bayes can be used to detect drowsiness by taking into account the probability of a given input being drowsy or not. By using Naive Bayes, we had achieved 69.18% accuracy.

Modified SVMs: Support Vector Machines are the supervised learning algorithms used for both classification and regression. SVMs are effective in detecting drowsiness since they can accurately separate the data points belonging to different classes. Modified SVMs are a variation of regular SVMs that are optimized for specific tasks. Modified SVMs are especially useful in detecting drowsiness since they can be trained to better recognize patterns associated with drowsiness. Modified SVMs can be used to generate a model that can detect drowsiness with a higher degree of accuracy than regular SVMs. The SVM classifier is trained to classify images as drowsy or non-drowsy based on the training data. Modified SVM produce an accuracy of 95.20% making it best suitable algorithm for detecting drowsiness.

Decision Trees (DTs): Decision Trees are supervised machine learning algorithms used for both classification and regression. Decision trees can be used to detect drowsiness since they can accurately separate the data points belonging to different classes. Decision Trees are used to generate a model that can detect drowsiness with a 88.70% of accuracy.

Random Forests: Random Forests are an ensemble learning algorithm that combines multiple decision trees and outputs the most frequent class. They can provide an accuracy of 92.46% for the provided data.

Logistic Regression: Logistic Regression is a supervised machine learning algorithm that can be used for both binary and multiclass classification. Logistic regression can be used to detect drowsiness since it can classify the data points into different classes based on the input features. Logistic regression was able to produce 93.49% accuracy.

In conclusion, After evaluating several algorithms for the task of detecting drowsiness, it was found that the modified Support Vector Machines (SVMs) algorithm had the highest accuracy at 95.20%. These results suggest that Modified SVMs are the most suitable algorithm for detecting drowsiness.

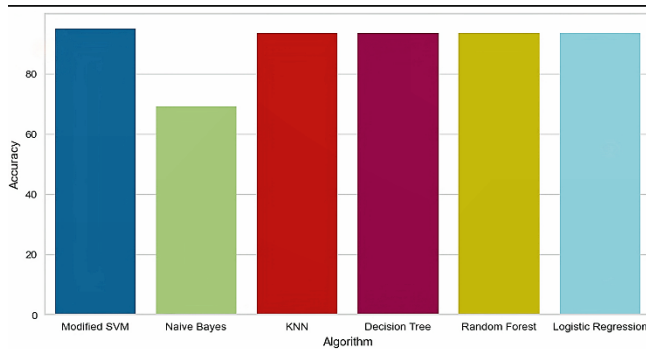


Fig. 6. Accuracy of each algorithm

B. Real-Time Monitoring

In Real-time monitoring we continuously monitor the driver's alertness and alert them if they show signs of drowsiness while driving. Real-time monitoring can be done, by capturing video frames from a camera that is mounted inside the vehicle and then processing these frames to extract relevant information about the driver. We use a variety of methods to detect drowsiness, such as monitoring eye movements, monitoring the driver's yawning or analyzing the driver's facial expressions. If drowsiness is detected, the device can alert the driver through auditory or visual cues to help them stay awake and alert.

The proposed system's framework is shown in Figure 7.

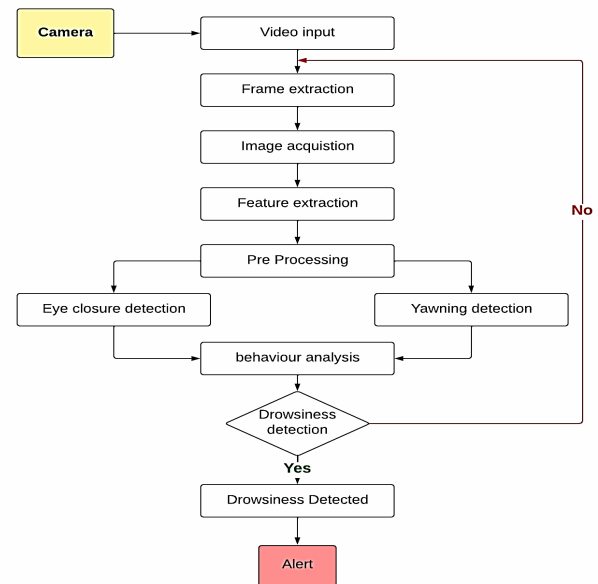


Fig. 7. System Architecture

1) Video Input:

This can be done using video cameras or other video capture devices, which can be mounted on the driver's body or in the vehicle.

In our work, the video input was captured using a webcam, which is a type of digital video camera that is typically connected to a computer or other device through a USB port. The webcam was used to capture a streaming video of the driver, which was then processed in real-time to detect signs of drowsiness and it allows continuous monitoring of the driver's alertness levels.

2) Frame Extraction:

In this Frame extraction, the video stream is taken and divided at rate of 20 frames per second (fps), then each frame represents a snapshot of the driver's appearance at that moment in time. By dividing the video stream into individual frames at a rate of 20 fps, the system can analyze each frame separately to detect signs of drowsiness, such as yawning and blinking.

3) Feature Extraction:

The frames are converted from the default BGR color space to the grayscale color space. The grayscale frame is then passed to the face detection function, which uses the pre-trained Haar Cascade classifier to detect faces in the image.

The images are processed and enhanced by reducing the noise and background blur using Gaussian blur technique. Finally, the convexHull is used to detect the coordinates of the eyes and mouth. These coordinates are used to draw contours around the features.

Once the location of the eyes and mouth has been detected, dlib can be used to extract specific features from these regions, such as the size and shape of the eyes or the movement of the mouth. These features are used as input to machine learning algorithms that can analyze the data to detect signs of drowsiness, such as yawning and blinking. This approach to feature extraction can be effective and

helpful in improving the accuracy of drowsiness detection systems.

4) Eye Closure Detection:

In this approach, real-time video streams are used to identify eye closure using the Eye Aspect Ratio (EAR). It measures the distances between specific spots on the eye, such as the angle between the outer and inner corners of the eye and the distance between the outer and center corners of the eye (horizontally). The eye can be considered to be closing if the ratio of these distances is below a predetermined threshold. This can be monitored over time to determine when the eye is entirely closed and to alert the driver if necessary.

5) Yawning Detection:

By monitoring the mouth's aspect in a video stream, we can identify drowsy driving using the Mouth Aspect Ratio (MAR). The ratio of the distance between the top and bottom lips to the distance between the outer corners of the mouth is used to compute the MAR. By tracking the MAR over time, it is possible to detect when the mouth is closing and when it is fully closed.

The system would need to track the location of certain points on the mouth in each frame of the video. Once the locations of the relevant points on the mouth have been determined, the MAR ratio can be calculated for each frame. If the MAR ratio falls below a certain threshold, this can be taken as an indication that the driver is drowsy.

6) Drowsiness Detection:

By tracking the EAR and MAR over time, the system is able to detect when the eye and the mouth are closing. The EAR and MAR ratios are calculated for each frame. If either ratio falls below a certain threshold, this can be taken as an indication that the driver is drowsy.

Immediately, the system will display a visual or audio alert. The visual alert is a message displayed on a screen, while the audio alert is a beep. By alerting the driver to their drowsy state, the system can help prevent accidents caused by drowsy driving.

IV. EXPERIMENTAL RESULTS

The experiment is conducted to detect drowsiness using Mouth Aspect Ratio (MAR) and Eye Aspect Ratio (EAR). It results the information about driver's eyes were closed or opened and number of times the driver yawned.

TABLE II. Here are the results section that included tabulation of the accuracy, precision, recall and F1-Score of the Algorithms.

Model	Precision	Recall	F1-Score	Accuracy
Modified SVM	0.96	0.95	0.95	0.952
Naïve bayes	0.71	0.69	0.68	0.691

KNN	0.83	0.81	0.80	0.818
Decision Tree	0.90	0.90	0.91	0.900
Random Forest	0.93	0.93	0.93	0.931
Logistic Regression	0.94	0.93	0.93	0.934

Here are some examples of the outcomes produced by our proposed system. Figures 8, 9, and 10 depict the various states of driver alertness detected by the system.

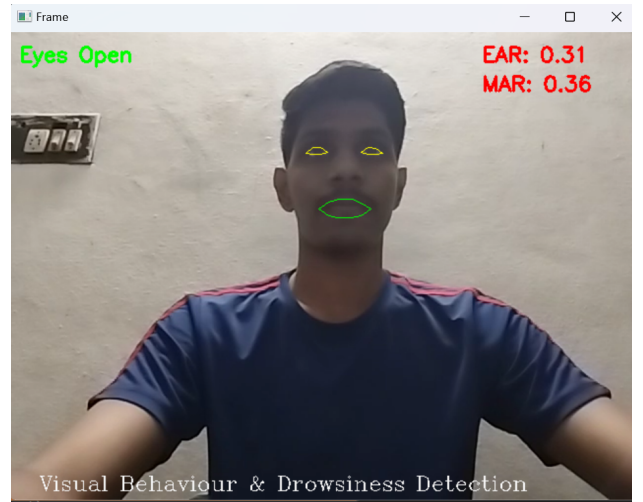


Fig. 8. Eyes open and Mouth closed

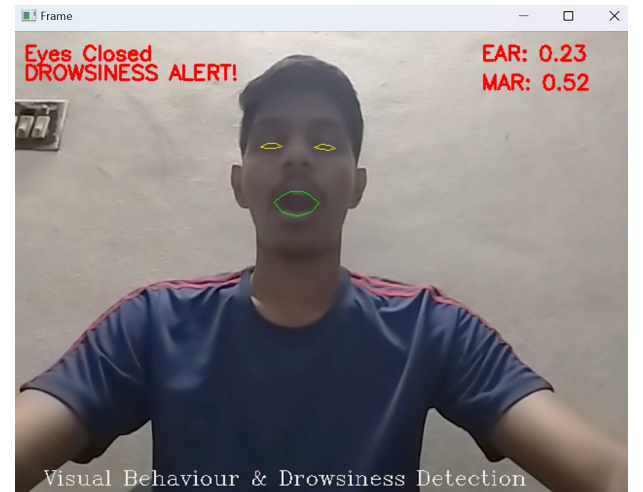


Fig. 9. Eyes closed and Yawning

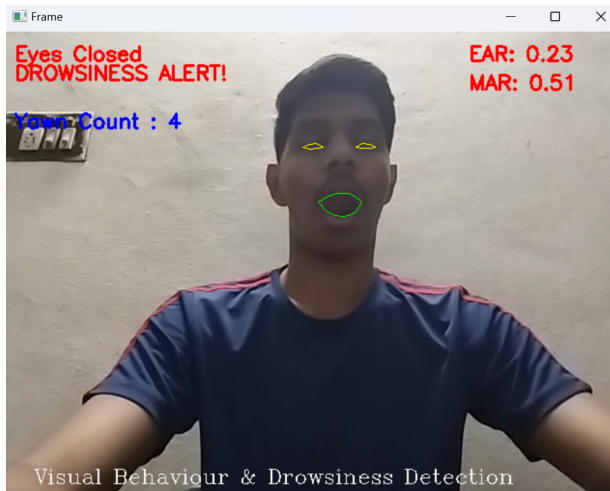


Fig. 10. Eyes closed and Yawning count

V. CONCLUSION

In this study, we investigated the effectiveness of real-time drowsiness detection techniques in various lighting conditions. We implemented and compared multiple machine learning algorithms and image processing clustering techniques, and found that the modified support vector machine (SVM) algorithm had the highest accuracy at 95.20%. Our system was able to detect drowsy driving by analyzing the driver's facial expressions and triggering an alarm when the Mouth Aspect Ratio (MAR) and Eye Aspect Ratio (EAR) fall below a certain threshold. The proposed algorithm performed well at an optimal distance from the light source, but its efficiency decreased with increasing distance and decreasing light. This system has the advantage of not requiring any special devices or sensors to be worn by the driver and can be implemented using a camera or any other imaging system mounted in the vehicle.

VI. FUTURE ENHANCEMENT

The system was designed with a focus on real-time monitoring, and emphasized flexibility, adaptability, and the ability to be enhanced. There are several potential areas for future enhancement of the system.

One possibility is to develop the system as a standalone product that can be installed in an automobile to monitor the driver. This would allow the system to be used in a variety of different vehicles and would not require the driver to have a separate device. Another option is to implement the system as a smartphone application that can be installed on a driver's smartphone. The driver could start the application and place the smartphone in a position where the camera is focused on their face, allowing the system to monitor their facial expressions and detect signs of drowsy driving. This approach would be convenient and portable, as the driver would only need to have their smartphone with them to use the system.

They have the potential to significantly improve the effectiveness and convenience of the system for detecting drowsy driving.

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